Model Development Phase Template

Date	25 February 2026
Team ID	259422
Project Title	Online Payments Fraud Detection using Machine Learning
Maximum Marks	4 Marks

Initial Model Training Code, Model Validation and Evaluation Report

The initial model training was performed using supervised machine learning algorithms on the preprocessed dataset. The dataset was split using a stratified 80:20 train-test split to preserve fraud class distribution.

The following models were trained:

Decision Tree Classifier

Random Forest Classifier

Extra Trees Classifier

All models were implemented using the scikit-learn Python library..

Initial Model Training Code:

```
#MODEL 1: Decision Tree
dtc=DecisionTreeClassifier()
dtc.fit(x_train, y_train)

y_test_pred_dtc=dtc.predict(x_test)
print('Decision Tree Classifier Test Accuracy:', accuracy_score(y_test, y_test_pred_dtc))

y_train_pred_dtc=dtc.predict(x_train)
print('Decision Tree Classifier Train Accuracy:', accuracy_score(y_train, y_train_pred_dtc))

print('Decision Tree Classifier Confusion Matrix:\n', pd.crosstab(y_test, y_test_pred_dtc))
print('Classification Report:\n', classification_report(y_test, y_test_pred_dtc))
```

```
#MODEL 2: Extra Tree Classifier
etc=ExtraTreesClassifier()
etc.fit(x_train, y_train)

y_test_pred_etc=etc.predict(x_test)
print('Extra Trees Classifier Test Accuracy:', accuracy_score(y_test, y_test_pred_e

y_train_pred_etc=etc.predict(x_train)
print('Extra Trees Classifier Train Accuracy:', accuracy_score(y_train, y_train_pre

print('Extra Trees Classifier Confusion Matrix\n:', pd.crosstab(y_test, y_test_pred

print(classification_report(y_test, y_test_pred_etc))
```

```
#MODEL 3: Random Forest Classifier
rfc=RandomForestClassifier()
rfc.fit(x_train, y_train)

y_test_pred_rfc=rfc.predict(x_test)
print('Random Forest Classifier Test Accuracy:', accuracy_score(y_test, y_test_pred_rfc))

y_train_pred_rfc=rfc.predict(x_train)
print('Random Forest Classifier Train Accuracy:', accuracy_score(y_train, y_train_pred_rfc))

print('Random Forest Classifier Confusion Matrix:\n', pd.crosstab(y_test, y_test_pred_rfc))

print(classification_report(y_test, y_test_pred_rfc))
```

Model Validation and Evaluation Report:

Model	Classification Report	Accuracy	Confusion Matrix
#MODEL 1: Decision Tree Classifier	Classification Report: precision recall f1-score support Fraud Transaction 0.90 0.89 0.89 1643 Legit Transaction 1.00 1.00 1.00 1270881 accuracy 1.00 1272524 macro avg 0.95 0.94 0.95 1272524 weighted avg 1.00 1.00 1.00 1272524	Decision Tree Classifier Test Accuracy: 0.999725741911351	Decision Tree Classifier Confusion Matrix: col_0 Fraud Transaction Legit Transaction isFraud Fraud Transaction 1459 184 Legit Transaction 165 1270716
#MODEL 2: Extra Tree Classifier	precision recall f1-score support Fraud Transaction 0.99 0.78 0.87 1643 Legit Transaction 1.00 1.00 1.00 1270881 accuracy 1.00 1272524 macro avg 0.99 0.89 0.94 1272524 weighted avg 1.00 1.00 1.00 1272524	Extra Trees Classifier Test Accuracy: 0.9997021667174844	Extra Trees Classifier Confusion Matrix : col_0 Fraud Transaction Legit Transaction isFraud Fraud Transaction 1279 364 Legit Transaction 15 1270866
#MODEL 3: Random Forest Classifier	precision recall f1-score support Fraud Transaction 0.97 0.79 0.87 1643 Legit Transaction 1.00 1.00 1.00 1270881 accuracy 1.00 1272524 macro avg 0.99 0.90 0.94 1272524 weighted avg 1.00 1.00 1.00 1272524	Random Forest Classifier Test Accuracy: 0.9997037383970754	Random Forest Classifier Confusion Matrix: col_0 Fraud Transaction Legit Transaction isFraud Fraud Transaction 1302 341 Legit Transaction 36 1270845